Group Presentations as Homotopy Pushouts

A Cubical Agda Formalization

Hongwei (Alex) Wang
New York University Abu Dhabi
hw4375@nyu.edu

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Abstract

For a presentation by a type of generators and a type of relations, we form a 1-truncated homotopy pushout obtained by attaching one circle for each relation. We construct the usual algebraic quotient of the free group by the normal closure of the relation differences and a canonical homomorphism from this group to the fundamental group of the resulting type. The principal formal result is that this homomorphism is surjective. The article gives a mathematical account of the checked Cubical Agda proofs; it does not assert injectivity, a presentation theorem for the fundamental group, or a classifying-space equivalence.

1 Introduction

Let G and R be types. A presentation with generators G and relations R may be realized geometrically by a graph with one loop for each generator, followed by the attachment of a circle for each relation. In the present construction the graph is expressed as a suspension, and the attachment is a homotopy pushout. The subsequent 1-truncation retains the range in which a fundamental group may be formed. The construction is carried out in Cubical Agda, using its higher inductive types for suspensions, circles, pushouts, and truncations [2, 1].

The input is universe-polymorphic: $G : \mathsf{Type}_\ell$, $R : \mathsf{Type}_{\ell'}$, and a presentation P. No sethood, decidability, or discreteness assumption is made on G or R. The source annotations below identify the declaration whose type and proof term establish the assertion in question. Environment and API details are deferred to Appendix A.

2 Presentations and loop words

Formal source. GroupPresentation.Presentation;GroupPresentation.Signed;GroupPresentation.Word;GroupPresentation.letterLoop;GroupPresentation.wordLoop;GroupPresentation.relationLoop₀;GroupPresentation.relationLoop₁;GroupPresentation.relationLoop.

Definition 2.1. *A* syntactic presentation P *on* G *and* R *consists of two functions* $p_0, p_1 : R \to \mathsf{Word}(G)$*, where a word is a finite list of signed generators. Once a based space carrying loops* ℓ_g *is fixed, a positive letter is evaluated by* ℓ_g*, a negative letter by* ℓ_g^{-1}*, and a word by concatenation. Write* $L_i(r)$ *for the loop represented by* $p_i(r)$ *and* $L(r) = L_0(r) \cdot L_1(r)^{-1}$ *for its relation-difference loop.*

The words are evaluated in a raw identity type, before set truncation. Thus the circle attachment uses the actual path $L(r)$ and requires no normal-form theorem for an untruncated loop space.

3 The generator space

Formal source. GroupPresentation.GeneratorInput;GroupPresentation.GeneratorSpace;GroupPresentation.generatorLoop.

Definition 3.1. *Let e denote the point of the auxiliary summand $\mathbf{1}$. The* generator space *is $X_G = \mathrm{Susp}(\mathbf{1} \sqcup G)$, based at its north pole. For $g : G$, put $\ell_g = \mathrm{merid}(g) \cdot \mathrm{merid}(e)^{-1}$. Thus each generator determines a north-based loop.*

Formal source. GroupPresentation.Connectedness.generatorSpaceMerePathToNorth.

Lemma 3.2. *Every point of X_G is merely equal to its north pole.*

Proof. The proof is suspension induction. At north it is represented by reflexivity, and at south by the reference meridian. Along a meridian the target is a proposition, namely a propositional truncation of an identity type; the two endpoint representatives therefore extend uniquely. □

Formal source. GroupPresentation.GeneratorBouquet.generatorSpace≃•Bouquet;GroupPresentation.GeneratorBouquet.generatorLoop-toBouquet;GroupPresentation.GeneratorBouquet.liftedGeneratorSpace≃•Bouquet.

Proposition 3.3. *There is a pointed equivalence $X_G \simeq_* \mathrm{Bouquet}(G)$ sending ℓ_g to the g-th bouquet loop; the same assertion holds after the universe lift used below.*

Proof. The forward suspension eliminator sends both poles to the bouquet base point, the e-meridian to reflexivity, and the g-meridian to the g-loop. The inverse sends the bouquet base point to north and its g-loop to ℓ_g. Their homotopies are respectively suspension and bouquet induction. The loop calculation follows by functoriality of concatenation and the unit law for the image of the reference meridian. The lifted proof is the same construction with G replaced by its lift. □

4 Relation circles and the presentation type

Formal source. GroupPresentation.attachingMap;GroupPresentation.relationProjection;GroupPresentation.PresentationPushout;GroupPresentation.BG_pres;GroupPresentation.BG-basepoint.

Definition 4.1. *For $r : R$, circle recursion sends the circle base point to north and its generating loop to $L(r)$. Let $a : R \times S^1 \to X_G$ be the resulting map and let $q : R \times S^1 \to R$ be projection. Define*

$$Y_P = \mathrm{Pushout}(a, q), \qquad B_P = \|Y_P\|_1.$$

The formal notation is PresentationPushoutP *and* BG_presP*; the latter is implemented as h-level truncation 3, the library's index for a 1-type.*

Formal source. GroupPresentation.Relations.relationCircleNaturality;GroupPresentation.Relations.conjugacyToRefl;GroupPresentation.Relations.inlRelationLoop-null;GroupPresentation.Relations.inlRelationSidesAgree.

Proposition 4.2. *For every $r : R$, the image of $L(r)$ in Y_P is null. Consequently the images of $L_0(r)$ and $L_1(r)$ in Y_P are equal.*

Proof. Write u for the pushout glue path at (r, base). Naturality of the family of glue paths around the circle gives $u = \mathsf{inl}(L(r)) \cdot u$. Cancelling u on the right proves $\mathsf{inl}(L(r)) = \mathsf{refl}$; the cited term performs this cancellation explicitly by the right unit, associativity, and inverse-cancellation laws. Since $L(r) = L_0(r) \cdot L_1(r)^{-1}$, a second cancellation calculation gives $\mathsf{inl}(L_0(r)) = \mathsf{inl}(L_1(r))$. □

Formal source. GroupPresentation.TruncatedRelations.generatorSpaceToBG;GroupPresentation.TruncatedRelations.BG-relationLoop-null;GroupPresentation.TruncatedRelations.BG-relationSidesAgree.

Corollary 4.3. *The canonical map $X_G \to B_P$ sends every relation loop to reflexivity and sends the two sides of each relation to equal loops.*

Proof. The map is the composite of the left pushout inclusion with the truncation constructor. Applying that constructor to the two preceding path equalities gives the asserted equalities in B_P. □

Formal source. GroupPresentation.Connectedness.presentationPushoutMerePathToBase;GroupPresentation.Connectedness.BG-pres-isConnected.

Proposition 4.4. *B_P is connected.*

Proof. Pushout induction reduces a point of Y_P either to a point of X_G, which is merely joined to north by the preceding lemma, or to a relation point, which is joined to the base point by the glue path at the circle base point. Truncation induction transports these merely inhabited path spaces to B_P. The final term contracts the set truncation at the selected base point, which is the library definition of isConnected2. □

5 The algebraic presented group

Formal source. GroupPresentation.PresentedGroup.PresentationFreeGroup;GroupPresentation.PresentedGroup.wordToFreeGroup;GroupPresentation.PresentedGroup.relationDifference;GroupPresentation.PresentedGroup.presentationNormalSubgroup;GroupPresentation.PresentedGroup.PresentedGroup.

Definition 5.1. *Put $F_P = \mathsf{FreeGroup}(\mathrm{Lift}_{\ell'} G)$. A signed word is evaluated in F_P, and $d_r = [p_0(r)][p_1(r)]^{-1}$. Let N_P be the normal subgroup generated by the lifted family $(d_r)_{r:R}$, and put $G_P = F_P/N_P$.*

The free group and quotient group are imported constructions. The normal closure is local: its raw generators are the d_r, the identity, products, inverses, and conjugates, followed by propositional truncation.

Formal source. GroupPresentation.PresentedGroup.NormalClosure.least;GroupPresentation.PresentedGroup.relationDifferenceInNormalClosure.

Lemma 5.2. *N_P is contained in every normal subgroup of F_P containing all d_r.*

Proof. Eliminate a truncated raw normal-closure witness. A relation generator is in the proposed subgroup by hypothesis; identity, product, and inverse are handled by the subgroup laws; and a conjugate is handled by normality. Hence the constructed subgroup is least. The particular inclusion of d_r is the raw relation constructor. □

6 The canonical homomorphism

Formal source. GroupPresentation.FundamentalGroup.BG-π₁;GroupPresentation.FundamentalGroup.relationLoopπ₁-null.

Let $\pi_1(B_P)$ denote the fundamental group at the designated base point. The relation loop has trivial class in this group, obtained by taking the set-truncation class of its nullhomotopy.

Formal source. GroupPresentation.PresentationHom.freeGroupToBG-π₁;GroupPresentation.PresentationHom.wordToFreeGroup-π₁;GroupPresentation.PresentationHom.relationDifferenceToBG-π₁-null;GroupPresentation.PresentationHom.presentationNormalSubgroupInKernel;GroupPresentation.PresentationHom.canonicalPresentationHom.

Proposition 6.1. *There is a canonical group homomorphism $c_P : G_P \to \pi_1(B_P)$.*

Proof. Send g to the class of the loop induced by ℓ_g. The upstream free-group recursor extends this to $F_P \to \pi_1(B_P)$. Induction on a signed word identifies its image with the corresponding concatenated loop class. Hence d_r maps to the class of $L_0(r)L_1(r)^{-1}$, which is trivial by the relation nullhomotopy. By normal-closure leastness, N_P lies in the kernel. The quotient eliminator therefore defines a map on F_P/N_P; quotient induction proves multiplicativity. This is c_P. □

7 Surjectivity

Formal source. GroupPresentation.BouquetFundamentalGroup.bouquetLoopHom-surjective;GroupPresentation.Surjectivity.inlPresentation↑-π₁-surjective;GroupPresentation.TruncationNaturality.truncationπ₁-surjective;GroupPresentation.CanonicalSurjectivity.freeGroupToBG-byTruncation≡freeGroupToBG-π₁;GroupPresentation.CanonicalSurjectivity.canonicalPresentationHom-surjective.

Theorem 7.1. *For every syntactic presentation P, the canonical homomorphism $c_P : G_P \to \pi_1(B_P)$ is surjective.*

Proof. The cited bouquet theorem constructs, for every loop class in a bouquet, a free-group word whose image is that class. Its proof uses the library maps windingT and loopingT: after passing through the inverse of the free-groupoid truncation equivalence, the local map bouquetLoopHom agrees with loopingT; the library homotopy left-homotopy makes the recovered word a section. The pointed equivalence of the generator-space section transports this surjectivity to the lifted X_G.

The left inclusion $X_G \to Y_P$ is surjective on fundamental groups. Indeed, the cited proof identifies every fibre of $R \times S^1 \to R$ with a lifted circle, proves that circle connected, and applies the upstream connected-pushout theorem followed by connectedFun→πSurj. The map $Y_P \to B_P$ induced by the truncation constructor is likewise surjective on fundamental groups: the local proof decodes a truncated loop with PathIdTruncIso, and the inverse law of that isomorphism gives a section.

Composing these three surjections gives a surjection from F_P to $\pi_1(B_P)$. On a free generator, its value is computed by path congruence as the class of the image of ℓ_g. Free-group homomorphism extensionality therefore identifies this composite with the previously defined free-group map underlying c_P. Finally a preimage in F_P is sent to its quotient class in G_P; propositional-truncation functoriality transports the equality of images. This is exactly the final proof term. □

8 Boundary of the result

The theorem establishes only surjectivity. No declaration proves that the kernel of c_P is trivial; hence neither $G_P \cong \pi_1(B_P)$ nor a Seifert–van Kampen description of this pushout follows from the

present development. In particular, although B_P is the natural candidate for a classifying type of the presented group, no pointed equivalence with a classifying space has been formalized. These are mathematical statements requiring additional proofs, not consequences of the notation BG_pres.

A Formal provenance and reproducibility

The local results used in the article are proved in GroupPresentation.Relations, GroupPresentation.TruncatedRelations, GroupPresentation.Connectedness, GroupPresentation.PresentedGroup, GroupPresentation.PresentationHom, GroupPresentation.GeneratorBouquet, GroupPresentation.BouquetFundamentalGroup, GroupPresentation.Surjectivity, GroupPresentation.TruncationNaturality, and GroupPresentation.CanonicalSurjectivity. The exact statement-to-source map, including upstream declarations, is maintained in alignment/records.jsonl.

The checked environment is Agda 2.8.0-3d04bac with the pinned Cubical checkout 87af036ebd1c87956cf08fea44e62c7cf76392ca, on its v0.9-compatible line. The relevant upstream ingredients are the suspension, pushout, truncation, free-group, quotient-group, bouquet fundamental-group, and connected-map interfaces of Cubical Agda. The local proofs cited above were checked with agda-isrc-icubical<module>; all source modules use {#- OPTIONS --cubical --guardedness #-}.

The proof of the main theorem uses, rather than re-proves, the upstream bouquet computations loopingT, windingT, and left-homotopy, the connected-map consequence connectedFun→πSurj, and the path/truncation equivalence PathIdTruncIso. Their roles are stated in the proof so that the mathematical argument remains readable independently of this implementation note.

References

[1] The Univalent Foundations Program. *Homotopy Type Theory: Univalent Foundations of Mathematics*. Institute for Advanced Study, 2013. https://homotopytypetheory.org/book/.

[2] Cubical Agda contributors. *Cubical Agda standard library*, pinned checkout 87af036ebd1c87956cf08fea44e62c7cf76392ca, v0.9-compatible line. https://github.com/agda/cubical.

[3] The Agda Team. *Agda 2.8.0*. https://agda.readthedocs.io/.